

Intermediate Quantitative Methods

What is a regression for?

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Week 2, part 1

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Today

- How to understand a regression?

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- How to understand a regression?
- Description

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- Description
- Prediction

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- Prediction
- Causal inference

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- How to understand a regression?
- Description
- Prediction
- Causal inference
- How the three get confused

How to understand a regression?

From last week

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- Last week the estimator was a sample mean
- This week it is a regression coefficient
- The harder question: which estimand is it an estimate of?

A regression

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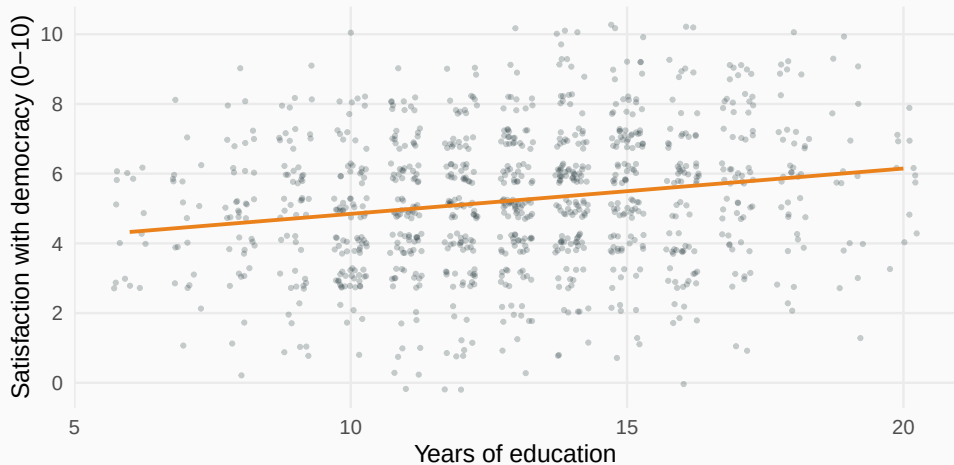
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- α : the intercept, the value of y when x equals zero
- ε : the error, everything else that moves y

Simulated survey, $n = 1,000$



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- Causal: one more year of education would raise a person's satisfaction by 0.13
- Obviously the meaning cannot come from the statistics themselves

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- “Differ by”, not “effect of”

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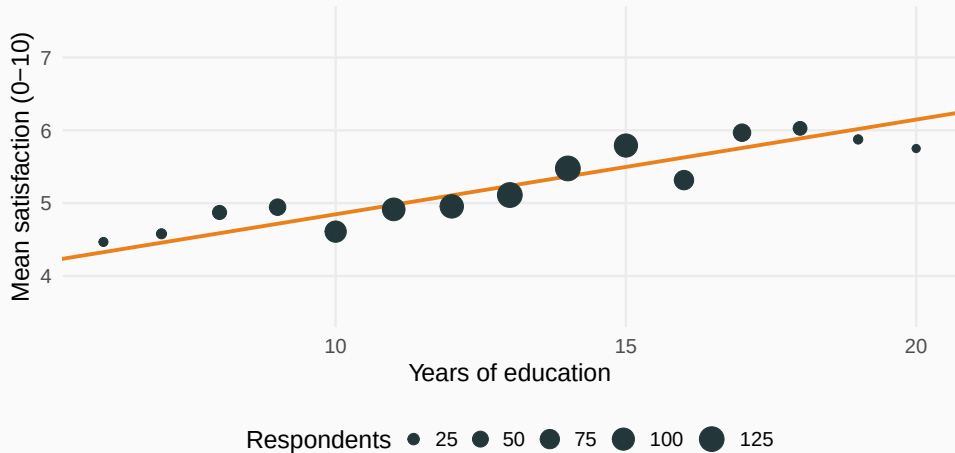
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- Binary x : β is exactly the difference between two group means
- Continuous x : the line smooths over many group means

Mean satisfaction at each year of education



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- Measures that capture the concepts
- A summary that fits the pattern: is a line a fair description?
- No assumption about why x and y move together

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- Gerring (2012): description treated as a lesser task
- Causal identification as the price of admission to top journals
- Descriptive findings contorted into causal framing to get published
- Result: a lot of weak causal claims, little careful description

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- Munger, Guess & Hargittai (2021)

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- Lührmann & Lindberg (2019): a “third wave of autocratization”, documented with V-Dem data

Prediction (including classification)

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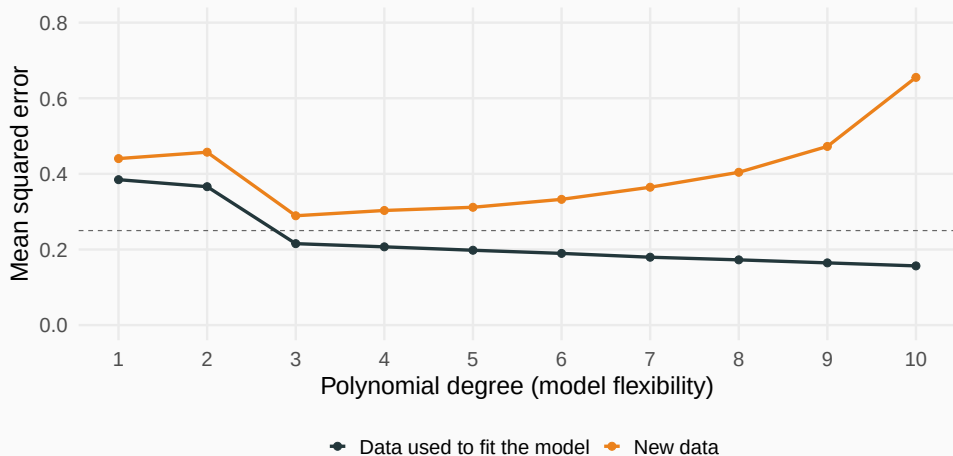
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- “ $\hat{y}$ problems” and “ $\hat{\beta}$ problems” (Mullainathan & Spiess 2017)

Error by model flexibility, 500 simulated datasets of 30



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- Week 8: machine learning

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- A model can identify a real effect and predict badly

Causal inference

A causal claim

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- β is causal only if nothing else that drives y differs along with x
- That condition comes from design and knowledge, not from the regression

A confounder: family background

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- The same children grow up with more economic security
- And with better experiences of schools, courts, public offices
- Both plausibly raise satisfaction with democracy in adulthood
- So the educated can be more satisfied even if schooling itself does nothing

- x : education, y : satisfaction with democracy, z : family background

Simulating it

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- z raises both x and y
- x has no effect on y at all

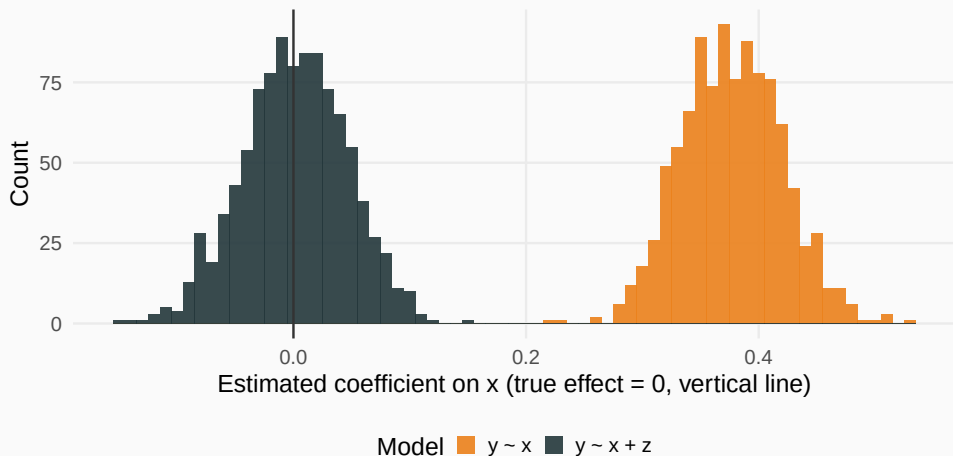
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- 1,000 datasets, two regressions on each

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- x : education, y : satisfaction with democracy, z : family background
- z raises both x and y
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- 1,000 datasets, two regressions on each
- $y \sim x$, then $y \sim x + z$

Estimates of an effect that is zero, 1,000 simulated datasets



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- Only the causal reading is wrong

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- Week 3: estimands and DAGs

How the three get confused

A typical regression table

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	Estimate	(SE)
Years of education	0.097	(0.019)
Age	0.012	(0.002)
Female	−0.100	(0.091)
Household income (decile)	0.161	(0.023)
Urban	−0.166	(0.092)
Born abroad	0.192	(0.141)
Intercept	1.475	(0.264)
<i>n</i>	2,000	
<i>R</i> ²	0.09	

Outcome: satisfaction with democracy (0–10). Simulated data.

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- The controls were never identified by the same argument
- Is 0.16 the effect of income? Is -0.17 the effect of living in a city?
- Westreich & Greenland (2013)

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- Test: is the description still interesting without the implied causal claim?

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- White Americans are older on average, and age drives COVID deaths
- Good description helps readers update their picture of the world correctly
- In today's US, age is always a confounder

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Causal inference	y if x changed	Design, assumptions	Confounding